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AI Deployment, Security & Compliance – Scenario-Based Quiz

Instructions

- Choose the **best answer (A/B/C/D)**
- Questions are based on **real-world situations**
- No direct theory recall – apply concepts

Section 1: AI Deployment Strategies

Q1

A startup deploys an AI chatbot directly on users' devices instead of the cloud to reduce latency and protect sensitive data.

What deployment strategy is being used?

- A. Centralized cloud deployment
- B. **Edge deployment**
- C. Hybrid deployment
- D. Multi-cloud deployment

Q2

A company runs its AI model in Azure but stores backup models in AWS for reliability. Which strategy is this?

- A. Edge computing
- B. **Multi-cloud deployment**

- C. On-premise deployment
- D. Serverless deployment

Q3

An AI fraud detection system must respond within milliseconds and cannot rely on internet connectivity.

Which deployment is best?

- A. Cloud deployment
- B. Edge deployment
- C. Hybrid deployment
- D. Batch processing

Q4

A healthcare company processes patient data locally but uses cloud AI models for predictions. What type of deployment is this?

- A. Hybrid deployment
- B. Edge-only deployment
- C. Centralized deployment
- D. Offline deployment

Q5

An AI model is deployed using containers and automatically scales based on traffic demand. Which concept is applied?

- A. Manual scaling
- B. Serverless architecture
- C. Containerized deployment with auto-scaling
- D. Static infrastructure



Section 2: Identity & Access Management (IAM)

Q6

A company ensures that developers can only access AI model logs but not modify the model. Which principle is being followed?

- A. Full access control
- B. Least privilege
- C. Open access
- D. Shared responsibility

Q7

An AI system requires users to log in using both password and OTP. What security method is this?

- A. Single sign-on
- B. Multi-factor authentication
- C. Role-based access
- D. Encryption

Q8

A cloud admin assigns permissions based on roles like "Data Scientist" and "Viewer". What model is this?

- A. Attribute-based access control
- B. Role-based access control (RBAC)
- C. Mandatory access control
- D. Open access control

Q9

An employee leaves the company, but their AI system access is not revoked. What risk does this create?

- A. Performance issue
- B. Security vulnerability
- C. Data redundancy
- D. Scalability issue

Q10

A system allows temporary access tokens instead of permanent credentials.
Why is this safer?

- A. Faster performance
- B. Reduced risk of credential misuse
- C. Easier implementation
- D. Lower cost



Section 3: Responsible AI Principles

Q11

An AI hiring system favors male candidates over female candidates due to biased training data.
Which issue is this?

- A. Overfitting
- B. Bias in AI
- C. Underfitting
- D. Data leakage

Q12

A bank wants to ensure its AI loan approval system explains why a loan was rejected.
Which principle is applied?

- A. Privacy
- B. Transparency
- C. Scalability
- D. Efficiency

Q13

An AI system is tested across different demographic groups before deployment.
What principle is being ensured?

- A. Security
- B. Fairness

- C. Latency
- D. Automation

Q14

An organization logs all AI decisions for audit purposes.
Which principle does this support?

- A. **Accountability**
- B. Scalability
- C. Optimization
- D. Redundancy

Q15

An AI model avoids using sensitive attributes like religion or caste.
What principle is followed?

- A. Bias amplification
- B. **Ethical AI**
- C. Data compression
- D. Feature engineering

A Section 4: Security, Privacy & Compliance

Q16

A company encrypts user data before storing it in the database.
What is this protecting?

- A. Data integrity
- B. **Data confidentiality**
- C. Data availability
- D. Data redundancy

Q17

An AI system masks personal data like phone numbers before processing.
What technique is this?

- A. Encryption
- B. Data anonymization
- C. Data replication
- D. Data compression

Q18

A company follows GDPR rules while handling European user data.
What does this ensure?

- A. Performance
- B. Compliance
- C. Scalability
- D. Automation

Q19

An attacker intercepts data between user and server.
Which solution prevents this?

- A. Firewalls
- B. Encryption (HTTPS/TLS)
- C. Load balancing
- D. Caching

Q20

A company regularly audits its AI systems for security vulnerabilities.
What practice is this?

- A. Debugging
- B. Compliance monitoring
- C. Data cleaning
- D. Model tuning



Section 5: Mixed Real-World Scenarios

Q21

An AI healthcare system stores sensitive patient data and must follow strict regulations. What is the top priority?

- A. Speed
- B. Security & compliance
- C. UI design
- D. Cost reduction

Q22

An AI app crashes when too many users access it at once. Which deployment improvement is needed?

- A. Data cleaning
- B. Auto-scaling
- C. Encryption
- D. Authentication

Q23

A company uses logs to track who accessed AI predictions and when. What does this ensure?

- A. Transparency
- B. Accountability
- C. Efficiency
- D. Redundancy

Q24

An AI system is trained on incomplete data and performs poorly in production. What issue occurred?

- A. Deployment failure
- B. Poor data quality

- C. Security breach
- D. Compliance violation

Q25

A company uses both IAM policies and encryption together.
What does this combination provide?

- A. Faster processing
- B. **End-to-end security**
- C. Reduced storage
- D. Simpler deployment